T-1: One-loop spectral gauge matching

A fixed-spectrum calculation in the four-copy sphere comparison

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Result and scope

The stated massive fermion spectrum produces an explicit, normalized one-loop contribution to the four-dimensional hypercharge kinetic coefficient. Its proper-time sum has both a power divergence and a logarithmic divergence. A zeta prescription computes a finite spectral remainder, but does not supply the finite local kinetic coefficient required to predict a physical gauge coupling.

The same spectrum gives a convergent one-loop contribution to the quartic hypercharge-field response:

$$\Delta\mathcal{L}^{\text{heavy fermions}}_{\mathscr{F}^4} = \frac{r^4}{1440\pi^2}\left(\frac{481}{72}+\frac{\zeta_R(3)}{54}\right)\left[(\mathscr{F}_{\mu\nu}\mathscr{F}^{\mu\nu})^2+\frac{7}{4}(\mathscr{F}_{\mu\nu}\widetilde{\mathscr{F}}^{\mu\nu})^2\right].$$

This is a Lorentzian effective-Lagrangian coefficient in the fixed connection convention below. It is a **heavy-fermion threshold contribution**, not the full photon scattering amplitude or a parameter-free measured fine-structure constant. Higher-dimensional local operators and other fields can add to it.

The calculation does not add particles, fluxes, a coupling-selection potential, or a defect. It takes the previously proposed four-copy sphere model as a conditional background. The global theory, physical vacuum, and phenomenological viability of that comparison model are not established by this note. The current original-corpus change log already distinguishes the electromagnetic normalization problem from an illustrative compactification [P1]. Historical sections that conflict with the corrected technical statement are not used as proofs [P2].

The new script passes **33 symbolic identities and 21 numerical checks**. The most recent saved 29-check equation/obstruction script was freshly rerun and passed. The earlier boundary-value solver and the complete historical test battery were **not** rerun in this pass.

1. Fixed background, fields, and normalization

The background is the unbroken product $\mathbb{M}_4 \times S^2_r$, with a round internal sphere of radius r, one unit of X monopole flux, and no internal color or hypercharge flux. Fermions have their minimal kinetic operators, with no bulk fermion masses, no Higgs condensate, and no additional Pauli or higher-derivative interactions. These restrictions separate this matching calculation from the nonuniform Higgs background and the wormhole calculations.

Use the conventional hypercharge connection $\mathscr{B}$:

$$D_\mu = \partial_\mu - iq\mathscr{B}_\mu + \cdots, \qquad q = \frac{y}{2}, \qquad \mathscr{F} = d\mathscr{B}.$$

Its four-dimensional kinetic term is $-\mathscr{F}^2/(4g_Y^2)$. The connection does **not** have canonical kinetic normalization until the value of g_Y is supplied. This convention differs by a factor of two from a connection multiplying y directly. The charges here are not the integral hypercharge labels $z = 3y$ used in the tensor-lattice audit.

There are four copies of the following repaired six-dimensional Weyl-fermion block [P3]. The geometric weak multiplicities come from the internal modes; they are not additional six-dimensional multiplicities in this table.

Field	Color dimension d_C	Monopole index $n = x$	y	$q = y/2$	6D chirality
Q	3	2	1/3	1/6	+
L	1	2	−1	−1/2	+
u	3	1	4/3	2/3	−
d	3	1	−2/3	−1/3	−
e	1	1	−2	−1	−
spectator	3	0	1/3	1/6	+

For a monopole index of magnitude n, the nonzero internal Dirac eigenvalues yield four-dimensional massive Dirac fields with

$$m_{p,n}^2 = \frac{p(p+n)}{r^2}, \qquad d_{p,n} = 2p + n, \qquad p = 1, 2, \ldots .$$

There are also n massless four-dimensional Weyl modes when $n > 0$. They remain in the low-energy theory and are not integrated out in the heavy threshold. The $n = 0$ formula reduces to the ordinary sphere spectrum $m_p = p/r$, with degeneracy $2p$ for each positive internal eigenvalue. The opposite internal eigenvalues pair into a four-dimensional Dirac mass; they must not be counted a second time [1–3].

Define the charge moments, including four copies and color but not the angular degeneracy:

$$w_n^{(k)} = 4 \sum_{i:\, x_i = n} d_{C,i} q_i^k .$$

The exact results are

n	$w_n^{(2)}$	$w_n^{(4)}$
0	1/3	1/108
1	32/3	176/27
2	4/3	7/27

The six-dimensional chirality sign does **not** multiply these parity-even loop contributions. Chirality-weighted anomaly cancellation therefore is not cancellation of these ordinary gauge-polarization terms. The determinant phase and anomaly-inflow problem are distinct from the parity-even magnitude calculated here.

2. The quadratic gauge term from the actual tower

For one four-dimensional Dirac field of charge q and mass m, the background-field proper-time calculation gives

$$\Delta(g_Y^{-2}) = \frac{q^2}{12\pi^2} \int_{\Lambda^{-2}}^{\infty} \frac{ds}{s} e^{-sm^2}.$$

This is the usual regulated charge-renormalization contribution [4]. For example, the quadratic term of $z \coth z = 1 + z^2/3 - z^4/45 + \cdots$ reproduces the coefficient in a constant magnetic background. The positivity of this cutoff contribution is not a statement about the sign of a separately subtracted finite remainder.

Set $t = s/r^2$ and $L = \Lambda r$. Define the **massive-only** heat sums

$$K_n(t) = \sum_{p=1}^{\infty} (2p + n) e^{-tp(p+n)}.$$

The fermionic threshold is

$$\boxed{\Delta(g_Y^{-2})_{\rm KK}(\Lambda) = \frac{1}{12\pi^2}\sum_{n=0}^{2} w_n^{(2)} \int_{L^{-2}}^{\infty} \frac{dt}{t} K_n(t).}$$

This is an action-normalization calculation, not the assignment of a geometric ratio to a coupling. A proper-time cutoff is a regulator; it has not been identified with a physical cutoff determined by the theory. Comparisons of proper-time, zeta, and dimensional prescriptions are important in extra-dimensional gauge matching [5,6].

2.1 Short-time expansion and zero-mode bookkeeping

Euler–Maclaurin expansion, equivalently expansion with Hurwitz zeta functions, gives

$$K_n(t) = \frac{1}{t} - \frac{n}{2} - \frac{1}{6} + \frac{5n^2-1}{60}t + \frac{21n^2-5}{1260}t^2 - \frac{7n^4-30n^2+7}{5040}t^3 + O(t^4).$$

The constant $-n/2$ is due to removing the n massless Weyl modes: each Weyl field counts as half a Dirac field in this parity-even loop normalization. It is not a flux-dependent change in the local leading gravitational heat coefficient.

For the complete massive fermion content,

$$K_Y(t) := \sum_n w_n^{(2)} K_n(t) = \frac{37}{3t} - \frac{157}{18} + \frac{203}{180}t + \frac{823}{3780}t^2 + \frac{509}{15120}t^3 + O(t^4).$$

Consequently,

$$\boxed{\Delta(g_Y^{-2})_{\rm KK} = \frac{1}{12\pi^2}\left[\frac{37}{3}L^2 - \frac{157}{18}\log L^2 + C_{\rm PT} - \frac{203}{180L^2} + O(L^{-4})\right].}$$

The omitted Weyl zero modes have charge-squared sum 40/3. Their four-dimensional fermion beta coefficient is $b_Y^{\rm fermion} = (2/3)(40/3) = 80/9$. In proper-time Dirac units their logarithm contributes 20/3 to the coefficient. Adding that back to $-157/18$ yields $-37/18$, the constant coefficient of the full six-dimensional fermion heat trace. This checks the ultraviolet bookkeeping without pretending the massless modes have a local inverse-mass expansion.

2.2 A finite spectral remainder in closed form

Let

$$\zeta_n(s) = \sum_{p=1}^{\infty} \frac{2p+n}{[p(p+n)]^s}, \qquad \Re s > 1,$$

and continue it analytically. For the three needed indices,

$$\zeta_n(0) = -\frac{n}{2} - \frac{1}{6},$$

$$\begin{aligned}
\zeta_0'(0) &= 4\zeta_R'(-1), \\
\zeta_1'(0) &= 4\zeta_R'(-1) - \frac{1}{2}, \\
\zeta_2'(0) &= 4\zeta_R'(-1) + 2\log 2 - 2.
\end{aligned}$$

Thus the weighted spectral derivative is

$$D_Y := \sum_n w_n^{(2)} \zeta_n'(0) = \frac{148}{3}\zeta_R'(-1) + \frac{8}{3}\log 2 - 8,$$

and the finite constant for the displayed hard proper-time lower limit is

$$\boxed{C_{\rm PT} = D_Y + \frac{157}{18}\gamma_E = -9.277780641643466176\ldots.}$$

The numerical value is a regulator-prescription remainder, not g_Y^{-2}. A negative remainder does not imply a negative physical kinetic energy.

At a general zeta normalization scale μ, the corresponding derivative contains

$$D_Y - \frac{157}{9}\log(\mu r).$$

Setting $\mu r = 1$ hides that logarithm; it does not determine the finite local counterterm. The normalization-scale ambiguity of zeta determinants is explicit in the heat-kernel formalism [7].

There is also a finite regularization trap: separately zeta-regularizing $\log p$ and $\log(p+n)$ with the weight $2p+n$ misses $-n^2/2$ in the derivative convention above. For $n = 1, 2$, that gives errors $-1/2$ and -2. The computation uses the full second-order eigenvalue and checks it independently by heat sums.

3. Why the absolute gauge coupling is still not selected

The power and logarithmic terms require local matching data. A six-dimensional covariant operator expansion includes, among other terms,

$$\mathcal{L}_{6,Y} \supset -\frac{1}{4}[Z_{Y,6} + c_R\mathcal{R}_6 + \cdots]\mathscr{F}_{MN}\mathscr{F}^{MN}.$$

In the connection convention used here, $[Z_{Y,6}] = \text{mass}^2$ and c_R is dimensionless. On the round product background, $\int_{S^2}\sqrt{h}\mathcal{R}_6 = 8\pi$. At the displayed operator orders, a renormalized matching equation therefore has the structure

$$\frac{1}{g_Y^2} = 4\pi r^2 Z_{Y,6}^{\rm ren} + 8\pi c_R^{\rm ren} + \Delta_{\rm KK}^{\rm ren} + \text{other fields, light running, and higher operators.}$$

The massive-only split carries its own light-mode subtraction. Its logarithm must not be interpreted as a standalone six-dimensional counterterm before the light contribution is reconciled. The full local ultraviolet heat coefficients determine the allowed counterterms [7].

A finite prescription change $\Delta_{\rm KK}^{\rm ren} \mapsto \Delta_{\rm KK}^{\rm ren} + c$, together with $c_R^{\rm ren} \mapsto c_R^{\rm ren} - c/(8\pi)$, leaves the total coupling unchanged. That is a scheme change. By contrast, freely changing the physical finite local coefficient while keeping the remainder fixed changes the theory. Neither freedom is eliminated by calculating the spectral remainder.

In particular, the previous kinetic counterfamily survives:

$$Z_{Y,6}^{\rm ren} \mapsto Z_{Y,6}^{\rm ren} + \varepsilon \quad\Longrightarrow\quad \Delta(g_Y^{-2}) = 4\pi r^2\varepsilon.$$

The one-loop fermion determinant in this fixed connection basis contains the charges and the kinetic spectrum, not an equation imposing a unique ε. This does not exclude a new independently justified ultraviolet boundary condition. It does exclude treating a choice of subtraction prescription as that physical condition. Power-sensitive matching without additional ultraviolet input is discussed explicitly in [5].

The calculation has not evaluated all quantum corrections. Bosonic, tensor, mixed-gauge, gravitational, and higher-loop contributions are outside this pass. Even an eventual cancellation of some divergent coefficients would not, on its own, prohibit the finite gauge-invariant Maxwell term or fix its value.

4. A convergent nonlinear hypercharge response

There is a positive calculational result that does not require discarding a divergent sum. For a massive four-dimensional Dirac field, the Lorentzian Euler–Heisenberg term is [4]

$$\Delta\mathcal{L}_{\mathscr{F}^4} = \frac{q^4}{1440\pi^2 m^4}\left[(\mathscr{F}^2)^2 + \frac{7}{4}(\mathscr{F}\widetilde{\mathscr{F}})^2\right].$$

Its angular sum involves

$$A_n = \sum_{p=1}^{\infty} \frac{2p+n}{p^2(p+n)^2}.$$

For $n > 0$,

$$\frac{2p+n}{p^2(p+n)^2} = \frac{1}{n}\left(\frac{1}{p^2} - \frac{1}{(p+n)^2}\right),$$

so the series telescopes. Hence

$$A_0 = 2\zeta_R(3), \qquad A_1 = 1, \qquad A_2 = \frac{5}{8}.$$

Combining the charge-fourth weights gives

$$\sum_n w_n^{(4)} A_n = \frac{1}{108}(2\zeta_R(3)) + \frac{176}{27} + \frac{7}{27}\frac{5}{8} = \boxed{\frac{481}{72} + \frac{\zeta_R(3)}{54}}.$$

This is the coefficient stated at the beginning of the report. Unlike the $\mathscr{F}^2$ sum, its massive-fermion series converges: its radial-level terms decay as p^{-3}.

4.1 A controlled tower-truncation estimate

If all three radial sums are truncated at $p = N$, the omitted dimensionless weighted sum is positive and obeys

$$0 < \delta A(N) \le \frac{733}{108N^2}.$$

For $n = 1, 2$ the tails are explicit finite telescoping expressions. For $n = 0$, the integral test bounds $2\sum_{p>N} p^{-3}$ by N^{-2}. This is a bound on the stipulated mathematical tower, not a bound on every possible additional ultraviolet interaction.

N	Partial weighted sum	Omitted tail	Stated upper bound
10	6.6468884949	0.0559273737	0.0678703704
30	6.6957613782	0.0070544904	0.0075411523
100	6.7021507769	0.0006650916	0.0006787037

The infinite sum is $6.7028158685770295238\ldots$. An effective theory cannot literally be trusted above its cutoff; this estimate shows how little of this particular finite threshold is contributed by the formal distant tail when a sufficiently large separation of scales exists.

4.2 What the finite coefficient does not predict

The background field is pure hypercharge, with external geometric weak fields set to zero. This is not the full photon amplitude. The geometric weak multiplets of excited fermions, mixed weak–hypercharge operators, Higgs breaking, and other field loops must be included before computing that amplitude.

For a canonically normalized hypercharge field, $\mathscr{B} = g_Y B_{\mathrm{can}}$, the displayed coefficient gains a factor g_Y^4. A finite spectral number therefore does not make the observable independent of the gauge normalization.

There are also allowed six-dimensional local $\mathscr{F}^4$ operators. Their renormalized coefficients add independent matching contributions; finiteness of the massive tower does not prove that those terms vanish. The result is a specific one-loop contribution of the assumed minimally coupled fermions. It is not the complete coefficient of the effective theory.

The derivative and weak-field expansion requires external momenta $|p_{\mathrm{ext}}|r \ll 1$, $|q\mathscr{F}|r^2 \ll 1$, and perturbative interactions in the energy range used. The massless Weyl modes remain dynamical; attempting to integrate them into this local Euler–Heisenberg expansion would be invalid.

5. Numerical and provenance checks

The finite-part calculation was checked in three distinct ways: a convergent Hurwitz continuation of each spectral zeta function; direct exponentially convergent sums of exponential integrals E_1; and independent quadrature of the heat kernel in logarithmic proper time. The Λr grid is $5, 10, 20, 50, 100$. The heat-integral comparison was performed at $5, 10, 20$.

After removing the explicitly displayed asymptotic terms through L^{-6}, the extracted proper-time constants differ from the closed expression by about 3.2×10^{-9} at $L = 5$ and about 1.3×10^{-11} or less on the higher-cutoff grid. These are finite numerical diagnostics, not rigorous floating-point enclosures. The lower-cutoff error includes the next asymptotic term, not merely rounding.

The quartic sum was independently checked using direct finite sums and exact Hurwitz/telescoping tails at $N = 5, 10, 30, 100, 300$. All finite checks passed. The report records exact rational charge coefficients; numerical values are not compared with a measured fine-structure constant or chosen to reproduce one.

The package includes the source script, result JSON, two CSV diagnostics, dependency versions, input hashes, and the unchanged most recent baseline script with its fresh log. Two source notes extracted from the supplied archives preserve which field table and kinetic convention were used. Neither the original archives nor the corpus were modified; nothing was sent to GitHub or Zenodo.

6. Disposition of the research aim

This calculation fills a specific missing step: it takes a defined charged spectrum through the loop determinant to normalized gauge-field operators. It distinguishes a divergent kinetic threshold, a prescription-dependent finite spectral remainder, and a convergent nonlinear threshold.

It does not establish the physical normalization of hypercharge, the complete electroweak photon, a primitive string, the globally completed tensor theory, a stable realistic compactification, or empirical confirmation. Those are not silently redefined as having passed because the computation ran.

The next ultraviolet proposal can now be tested against an explicit matching requirement: it must determine the finite local kinetic data in the same charge convention and renormalization scheme, not merely supply another geometric number. For a measurable nonlinear response, the next calculation would instead be full mixed electroweak matching with a stated ultraviolet operator budget. These are alternative research tasks, not conclusions of this note.

Appendix: analytic-continuation formula used for the cross-check

With $a = 1 + n/2$ and $u = n/2$, the massive spectrum gives

$$\zeta_n(s) = 2\sum_{j=0}^{\infty} \frac{(s)_j}{j!} u^{2j} \zeta_H(2s+2j-1, a).$$

The $j = 1$ term has a pole in $\zeta_H(1+2s, a)$ that cancels its explicit factor of s. Keeping it is essential. Differentiating at zero gives

$$\zeta_n'(0) = 4\zeta_H'(-1, a) - 2u^2\psi(a) + 2\sum_{j=2}^{\infty} \frac{u^{2j}}{j} \zeta_H(2j-1, a).$$

For $n = 0, 1, 2$ this converges rapidly and reproduces the stated closed forms. The code uses 55-decimal working precision and 128 tail terms. The three continuation residuals are below the stated 10^{-42} test tolerance. They check the finite formulas independently of the cutoff extraction.

References and source boundaries

[**P1**] Supplied *A Theory of Everything*, Part 1 of 4, opening claim-by-claim change log, research snapshot 19 September 2026, especially pages 7–11. The snapshot states that flux integers and heat ratios do not by themselves fix the photon kinetic coefficient, and that illustrative compactifications are not the corpus's action.

[**P2**] Supplied Parts 2–4, section-audit inserts: historical, interpretive, and superseded statements are not current proofs. This note does not reactivate their former grand-unification claims.

[**P3**] Conversation archive `t1_sphere_higgs_audit_2026-09-28.zip`, `RESEARCH_NOTE.md`, sections 1–3: the six-field repaired block. Conversation archive `t1_goal_completion_audit_2026-10-12.zip`, `RESEARCH_NOTE.md`, section 7: the conventional $y/2$ hypercharge normalization. Full-file hashes are in `INPUT_PROVENANCE.json`.

[**1**] R. Camporesi and A. Higuchi, *On the eigenfunctions of the Dirac operator on spheres and real hyperbolic spaces*, arXiv:gr-qc/9505009, especially the sphere eigenvalues and degeneracy formula (3.54). Standard spectral input, not a result of T-1.

[**2**] R. Jante and B. J. Schroers, *Dirac operators on the Taub-NUT space, monopoles and SU(2) representations*, arXiv:1312.4879. Used for the monopole Dirac kernel and its representation interpretation.

[**3**] H. Dohi, T. Kakuda, K. Nishiwaki, K. Oda and N. Okuda, *Notes on sphere-based universal extra dimensions*, arXiv:1406.1954, section 3.2, equations (49)–(67). Its independent weak gauge sector and phenomenological model are not identified with the comparison here; only the minimally coupled sphere spectrum is used.

[**4**] G. V. Dunne, *Heisenberg–Euler Effective Lagrangians: Basics and Extensions*, arXiv:hep-th/0406216, sections 1.2.3 and 1.3.1, equation (1.17). Used for charge renormalization and the one-loop spinor quartic electromagnetic-field term.

[**5**] J. F. Oliver, J. Papavassiliou and A. Santamaria, *Can power corrections be reliably computed in models with extra dimensions?*, arXiv:hep-ph/0302083, Phys. Rev. D 67, 125004 (2003). Used for the distinction between power-sensitive cutoff matching and a physical coupling prediction.

[**6**] D. M. Ghilencea, *Regularisation Techniques for the Radiative Corrections of Wilson lines and Kaluza–Klein states*, arXiv:hep-th/0311187. Used for the need to reconcile dimensional, zeta, and proper-time prescriptions; its orbifold spectra are not substituted for this sphere spectrum.

[**7**] D. V. Vassilevich, *Heat kernel expansion: user's manual*, arXiv:hep-th/0306138, especially equations (2.29)–(2.33) and the counterterm discussion. Used for the determinant/heat-kernel connection and the normalization ambiguity. The specific charge-weighted sums in this note are new calculations within the stipulated model, not quoted results from this reference.